

# Facial Emotion Recognition

A Comparative Study of Custom CNN, ResNet-50, and Vision Transformer (ViT-B/16)

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Custom CNN

ResNet-50

ViT-B/16

## Facial Emotion Recognition (FER)

Supervised 7-class image classification: Angry · Disgust · Fear · Happy · Neutral · Sad · Surprise



### Medical Diagnosis

Pain & mood assessment



### Driver Monitoring

Fatigue & distraction alerts



### Education

Engagement tracking



### Customer Analytics

Sentiment feedback

## Key Challenges

- Real-world images: pose, lighting, partial occlusion variation
- Annotation subjectivity — ambiguous states (fear vs. surprise)
- Severe class imbalance (happy 38.9% vs. fear 2.3%)

## Objectives

1. Design and implement a custom CNN trained from scratch as a baseline.
2. Fine-tune ResNet-50 to assess the benefit of residual learning and transfer learning.
3. Fine-tune ViT-B/16 to assess whether global self-attention outperforms local convolutional feature learning.
4. Quantitatively compare all three models, with emphasis on minority classes.

## From Hand-Crafted Features to Transformers

Pre-2013

### Hand-Crafted Features

#### LBP + HOG + Gabor

Classic descriptors combined with SVM classifiers

#### CK+ & JAFFE

Strong on controlled lab datasets (brittle on real-world dataset)

#### Limitation

Failed under pose, lighting & occlusion variation

2013 – 2020

### Deep CNNs

#### Goodfellow et al. (2013)

FER-2013 benchmark — established CNNs as the standard

#### He et al. (2016)

ResNet — skip connections solve deep network degradation

#### Huang et al. (2023)

Fine-tuning pretrained ResNet on RAF-DB: +27% over scratch

2021 – Present

### Vision Transformers

#### Dosovitskiy et al. (2020)

ViT — global self-attention from patch tokens

#### Li et al. (2021)

Mask-ViT for FER: 88.62% on RAF-DB

#### Mao et al. (2022)

AU-aware ViT: 91.10% on RAF-DB

# DATASET

## RAF-DB: Real-world Affective Faces Database

**15,339**

Total images (RGB, 100×100 px)

**7**

Emotion classes

**17×**

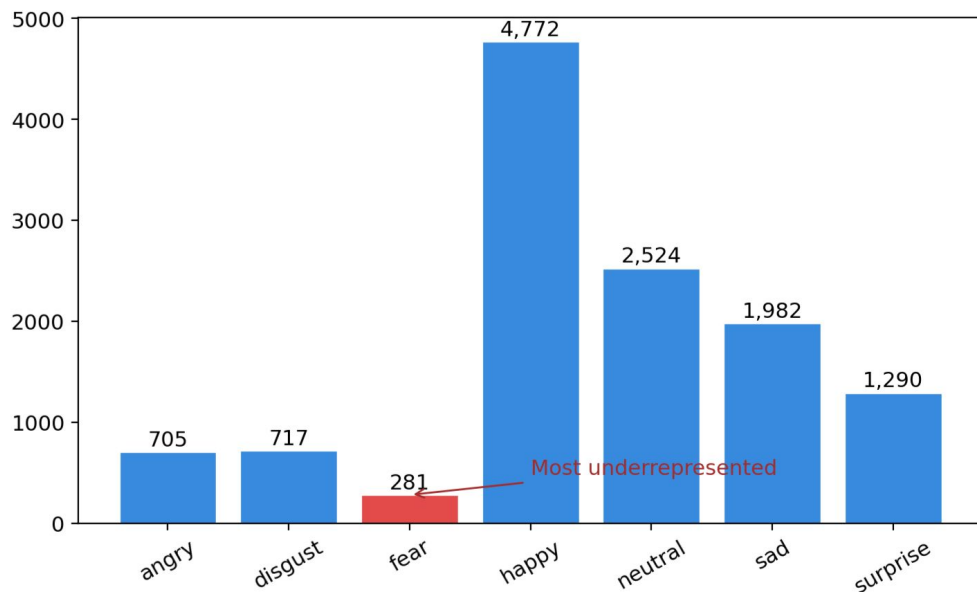
Class imbalance (Happy vs. Fear)

**~40**

Annotators per image



*Training Set Class Distribution  
(Fear = most underrepresented; Happy = dominant)*



## Preprocessing & Data Augmentation

### Preprocessing Pipeline

#### Resize

100×100 → 224×224 px (bilinear interpolation)

#### Normalize

Per-channel  $\mu=[0.575, 0.450, 0.401]$ ,  
 $\sigma=[0.206, 0.189, 0.180]$

#### Split

(seed=42)



Train: 11,044 (71.9%)



Validation: 1,227 (8.0%)



Test: 3,068 (20.0%)

Class weight:  $wc = N / (C \cdot nc)$   
where  $C=7$ ,  $nc$ =class count

### Augmentation (Training Only)

#### RandomHorizontalFlip

$p=0.5$  — bilateral face symmetry

#### RandomRotation

$\pm 12^\circ$  — head tilt variation

#### RandomCrop

224px, pad=10 (reflect) — positional shift

#### ColorJitter

brightness  $\pm 0.25$ , contrast  $\pm 0.25$ , saturation  $\pm 0.15$

#### RandomGrayscale

$p=0.1$  — prevent color-based shortcuts

# MODEL ARCHITECTURES

## Three Architectures — One Benchmark

All models trained identically: same loss, optimizer, data pipeline. Differences → architecture alone.

### Custom CNN

*Baseline (random init)*

**Parameters:** 456K

**Trainable:** 456K (all)

**Pretrained on:** None

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4 conv blocks (32→64→128→256 filters)  
+ GAP + FC (256) + Dropout (0.5)

### ResNet-50

*Residual learning*

**Parameters:** 23.5M

**Trainable:** 15.0M (Layer 4 + head)

**Pretrained on:** ImageNet (1.2M)

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Frozen: Stem + Layers 1–3  
Trainable: Layer 4 + 7-class head  
Skip connections for deep training

### ViT-B/16

*Global self-attention*

**Parameters:** 85.8M

**Trainable:** 21.3M (Blocks 10–12)

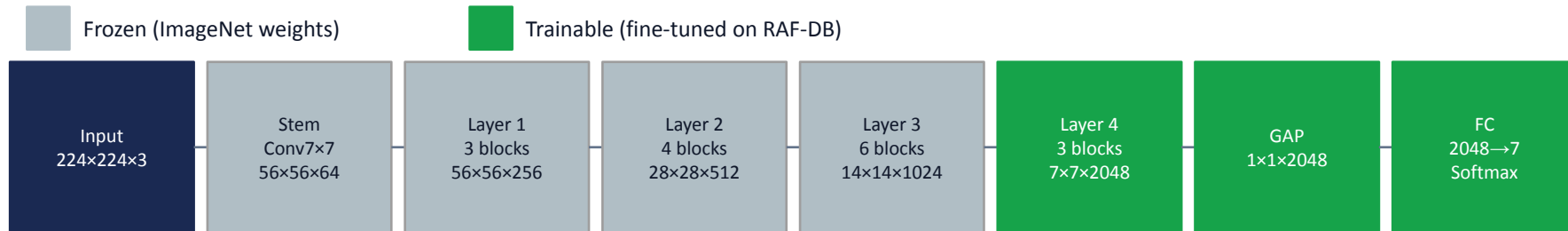
**Pretrained on:** ImageNet-21k (14M)

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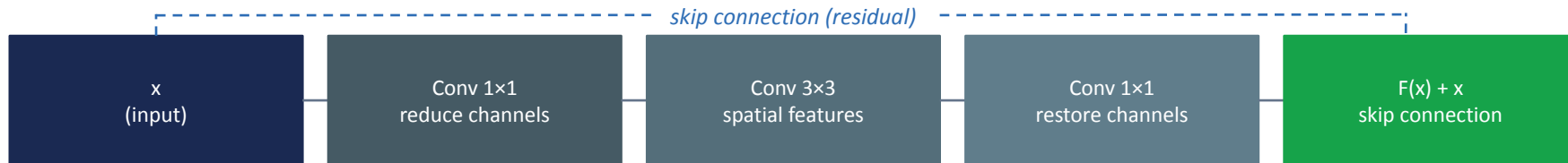
196 patches (16×16px) → 768-dim  
Frozen: Blocks 1–9  
Diff. learning rates: 1e-5 / 1e-4

# MODEL ARCHITECTURES

## ResNet-50: Residual Learning + Transfer Learning



### Bottleneck Block Detail



### Fine-Tuning Strategy

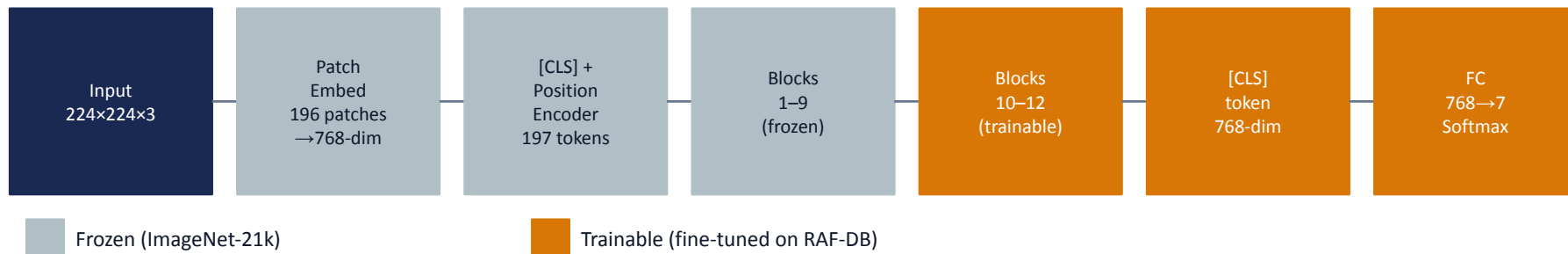
**Frozen:** 8.5M params (Stem + Layers 1–3)

**Trainable:** 15.0M params (Layer 4 + 7-class head)

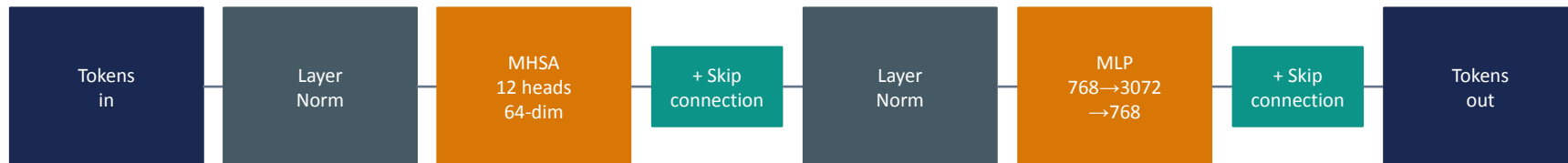
**Rationale:** Edge detectors & textures transfer directly to face images

# MODEL ARCHITECTURES

## ViT-B/16: Vision Transformer with Transfer Learning



### Transformer Encoder Block Detail (×12)



**CNN vs. ViT:** CNNs build spatial context hierarchically via local convolutions. ViT attends globally from Block 1 — every patch to every other patch. Differential LR:  $1e-5$  (Blocks 10–12) vs  $1e-4$  (head) prevents catastrophic forgetting.



# EXPERIMENTAL SETUP

## Training Configuration & Hyperparameters

| Property       | Custom CNN                                 | ResNet-50                                  | ViT-B/16                                   |
|----------------|--------------------------------------------|--------------------------------------------|--------------------------------------------|
| Learning rate  | 1e-3                                       | 1e-4                                       | 1e-5 / 1e-4                                |
| Optimizer      | Adam                                       | Adam                                       | Adam                                       |
| Batch size     | 32                                         | 32                                         | 32                                         |
| Max epochs     | 100                                        | 100                                        | 100                                        |
| LR scheduler   | ReduceLROnPlateau (factor 0.5, patience 5) | ReduceLROnPlateau (factor 0.5, patience 5) | ReduceLROnPlateau (factor 0.5, patience 5) |
| Early stopping | Patience=10, monitor val macro-F1          | Patience=10, monitor val macro-F1          | Patience=10, monitor val macro-F1          |
| Dropout        | 0.5                                        | —                                          | —                                          |
| Weight decay   | 1e-4                                       | 1e-4                                       | 1e-4                                       |

**Hardware:** NVIDIA A100 GPU (40 GB VRAM) on Google Colab

**Framework:** PyTorch + torchvision + HuggingFace Transformers

**Seed:** 42 (fully reproducible)

# EXPERIMENTAL SETUP

## Evaluation Metrics

Macro-F1 is the primary metric: it weights all 7 classes equally, unlike accuracy which is dominated by the happy class.

### Macro-averaged F1

PRIMARY

$$\text{Macro-F1} = (1/C) \sum_c 2 \cdot P_c \cdot R_c / (P_c + R_c)$$

F1 computed per class, then averaged equally. Primary metric.

### Overall Accuracy

$$\text{Acc} = \sum_c TP_c / N$$

Supplementary only — misleading under 17× class imbalance.

### Precision & Recall (per class)

$$P_c = TP_c / (TP_c + FP_c) \quad R_c = TP_c / (TP_c + FN_c)$$

Reported per class to diagnose minority-class performance.

### One-vs-Rest ROC-AUC

Area under TPR<sub>c</sub> vs FPR<sub>c</sub> as threshold varies

TPR<sub>c</sub> = TP<sub>c</sub> / (TP<sub>c</sub> + FN<sub>c</sub>), FPR<sub>c</sub> = FP<sub>c</sub> / (FP<sub>c</sub> + TN<sub>c</sub>). Macro avg reported.

**Confusion Matrix:** C×C matrix where entry (i, j) = # samples of true class i predicted as class j.

## Training Dynamics: Loss Curves & Convergence



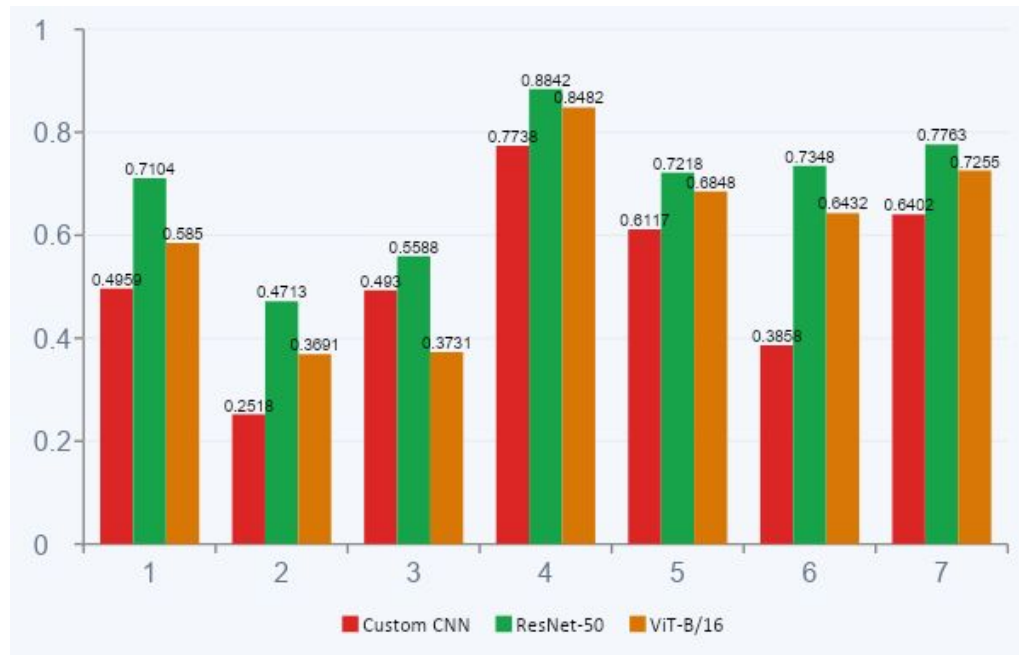
### Key Observations

- Custom CNN:** Slow oscillatory convergence over 97 epochs. Train  $\approx$  Val loss  $\rightarrow$  underfitting. Random init insufficient for this dataset size.
- ResNet-50:** Sharp train loss drop in first 10 epochs. Val loss plateaus  $\sim 1.4 \rightarrow$  slight overfitting in Layer 4. Best macro-F1 at epoch 40.
- ViT-B/16:** Fastest convergence (both losses fall steeply in first 5 epochs). Val loss stable  $\sim 0.95$  from epoch 10. Largest val $\rightarrow$ test macro-F1 drop (0.6517  $\rightarrow$  0.6041).

# RESULTS

## Per-Class F1 Score Comparison

| Emotion  | CNN    | ResNet   | ViT    |
|----------|--------|----------|--------|
| Angry    | 0.496  | 0.710    | 0.585  |
| Disgust  | 0.252  | 0.471    | 0.369  |
| Fear     | 0.493  | 0.559    | 0.373  |
| Happy    | 0.774  | 0.884    | 0.848  |
| Neutral  | 0.612  | 0.722    | 0.685  |
| Sad      | 0.386  | 0.735    | 0.643  |
| Surprise | 0.640  | 0.776    | 0.726  |
| Macro    | 0.5217 | 0.6939 ★ | 0.6041 |



*ResNet-50 achieves best F1 in every class. Disgust = hardest across all models. Happy = easiest across all models.*

# RESULTS

## ROC-AUC Scores (One-vs-Rest)

| Emotion   | Custom CNN   | ResNet-50    | ViT-B/16     |
|-----------|--------------|--------------|--------------|
| Angry     | 0.910        | <b>0.958</b> | 0.939        |
| Disgust   | 0.766        | <b>0.897</b> | 0.887        |
| Fear      | 0.906        | <b>0.928</b> | 0.911        |
| Happy     | 0.898        | <b>0.969</b> | 0.956        |
| Neutral   | 0.871        | <b>0.937</b> | 0.921        |
| Sad       | 0.829        | <b>0.951</b> | 0.919        |
| Surprise  | 0.922        | <b>0.969</b> | 0.964        |
| Macro AUC | <b>0.872</b> | <b>0.944</b> | <b>0.928</b> |



Macro-AUC ranking mirrors macro-F1: ResNet-50 (0.944) > ViT-B/16 (0.928) > Custom CNN (0.872). Disgust = lowest AUC in all models.

# RESULTS

## Overall Test-Set Performance

Macro-F1 is the primary metric (accounts for 17× class imbalance)

**77.18%**

ResNet-50 Accuracy  
(best model)

**0.6939**

ResNet-50 Macro-F1  
(primary metric)

**0.944**

ResNet-50 Macro-AUC  
(one-vs-rest)

| Model            | Accuracy        | Macro-F1 (test) | Macro-Precision | Macro-Recall  | Macro-AUC      |
|------------------|-----------------|-----------------|-----------------|---------------|----------------|
| Custom CNN       | 61.54%          | 0.5217          | 0.5241          | 0.5335        | 0.872          |
| <b>ResNet-50</b> | <b>77.18% ★</b> | <b>0.6939 ★</b> | <b>0.6938</b>   | <b>0.6969</b> | <b>0.944 ★</b> |
| ViT-B/16         | 70.96%          | 0.6041          | 0.5865          | 0.6333        | 0.928          |

★ Best in class | All three models trained under identical conditions

# DISCUSSION

## CNN → ResNet-50 Gap

- 50-layer residual depth vs. 4-block baseline
- Skip connections prevent vanishing gradients
- ImageNet pretraining transfers edge detectors & texture filters directly to face images

*macro-F1: 0.5217 → 0.6939 (+0.172)*

## ResNet-50 vs ViT Gap

- ViT lacks spatial inductive bias (translation equivariance, locality)
- ViT needs far more data to learn spatial structure (11K images insufficient)
- ViT shows largest val→test drop (0.6517 → 0.6041): mild overfitting

*macro-F1: 0.6939 vs. 0.6041*

## Disgust: Hardest Class

- 717 training samples — NOT sample scarcity
- Confusion matrices: disgust → happy (0.21) and → neutral (0.22)
- Intrinsic visual ambiguity with neighboring emotional states
- Class-weighted loss cannot resolve this inter-class overlap

*best F1 = 0.4713 (ResNet-50)*

## Limitations & Next Steps

Dataset size (15K) limits ViT · Generative oversampling for disgust/fear · Cross-dataset generalization

# Conclusion

01

## Custom CNN

61.54% accuracy · macro-F1 0.5217

Underfitting confirmed (random init insufficient at this dataset scale)

02

## ResNet-50

77.18% accuracy · macro-F1 0.6939

Residual learning + ImageNet transfer = strongest performer across all metrics

03

## ViT-B/16

70.96% accuracy · macro-F1 0.6041

Global self-attention offers NO advantage over CNNs at RAF-DB's scale (11K images)

04

## Disgust class

Best F1: 0.4713 (ResNet-50)

Intrinsic visual ambiguity (not solvable by class weighting alone)